

Risques climatiques — M2 Actuariat

# Optimal climate policy and the social cost of carbon

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Gauthier Vermandel

Sorbonne Université — M2 Actuariat

Wednesday, 16 September 2026, 08:30–10:30

## Objectives

- Formulate the social planner's **optimal control** problem (Ramsey setup).
- See how the **SCC emerges as a shadow price** and maps to an optimal carbon tax (USD/tCO<sub>2</sub>).

## Materials:

- DICE\_Notebook2\_optimization.ipynb *notebook exercises about scenarii simulations.*
- DICE\_Notebook2-2\_discounting\_tipping.ipynb notebook bonus on uncertainty.
- Handout\_DICE.pdf a synthetic presentation of the DICE model.

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# Introduction: From what could happen to what should happen

- In the previous lecture: we built **transition scenarios** (SSPs, forcing pathways) — describing *possible futures*.
- But in principle, we may also ask: **what should the transition look like?**
  - Formalized as an **optimal control problem**:

$$\max_{\{\mu_t, \dots\}} \sum_{t=0}^{\infty} \beta^t U(c_t) \quad \text{s.t. climate-economy dynamics}$$

- Find the sequence of decisions that maximizes social welfare.
- Advantages:
  - Reduces an infinite set of possible paths to one “optimal” path.
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- The key policy metric is not the abatement rate itself, but the **optimal carbon price** (or **Social Cost of Carbon, SCC**).
- Definition: the SCC is the **monetary value (USD per ton of CO<sub>2</sub>)** of the marginal damage from one additional ton of emissions.
- Why it matters:
  - Abstract concept (model-based), but provides guidance on what carbon price should be today.
  - Forms the basis for actual carbon tax design.
  - Example: in the US, the Biden Administration used five IAM-based SCC estimates to guide federal policy.

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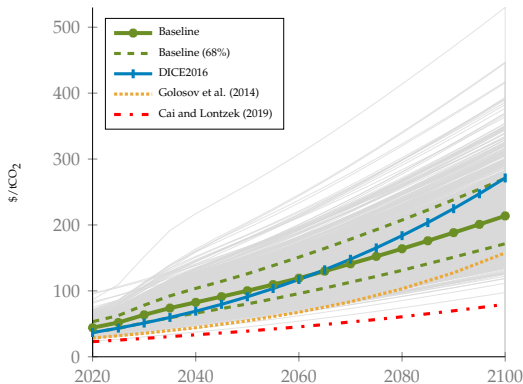
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# Introduction: Example



Note: The figure displays the social cost of carbon (SCC) on the vertical axis. The baseline model from [Jondeau et al. \(2022\)](#) is represented by a solid green line. Uncertainty interval, computed from 2,000 simulations using alternative parameter draws from the Metropolis-Hastings algorithm, are depicted by solid gray lines. Alternative SCC are also presented: (i) DICE2016 (solid blue with vertical markers), (ii) [Golosov et al. \(2014\)](#) (orange dotted line), and (iii) [Cai and Lontzek, 2019](#) with a benchmark calibration assuming 1.5% total factor productivity growth (red dashed line).

- 1 Introduction
- 2 The Optimal Control Problem
- 3 The SSC in the Ramsey Problem
- 4 Recap

# The Optimal Control Problem

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## Control Problem: the objective function

- **Individual preferences.** Utility from consumption  $u(c)$  with  $u'(c) > 0$  and  $u''(c) < 0$  (diminishing marginal utility). We use CRRA:

$$u(c) = \begin{cases} \frac{c^{1-\gamma} - 1}{1-\gamma}, & \gamma \neq 1, \\ \log c, & \gamma = 1, \end{cases} \quad \gamma > 0.$$

- **Aggregation and units.** Let  $C_t$  be aggregate consumption (trillion USD) and  $L_t$  population (million). Per-capita consumption in *thousand USD* is

$$c_t = 1,000 \times \frac{C_t}{L_t}.$$

Instantaneous social welfare is  $L_t u(c_t)$ .

- **Intertemporal welfare.** With pure rate of time preference  $\rho > 0$ ,

$$W_t = \sum_{i=0}^{\infty} \frac{L_{t+i} u(c_{t+i})}{(1+\rho)^{\Delta i}}.$$



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## Control Problem: the solution toolkit

**Planner's problem.** Consider a government that chooses decisions variables (savings/investment  $S_t$  and abatement  $\mu_t$ ) to maximize

$$\max_{\{s_t, \mu_t\}_{i \geq 0}} \sum_{i=0}^{\infty} \frac{L_{t+i} u(c_{t+i})}{(1 + \rho)^{\Delta i}} \quad \text{s.t. resource, capital, ...}$$

How to solve the problem?

1. **Numerical solver:** use solver to find sequences  $\{S_t, \mu_t\}_{t=0}^T$  that maximizes objective.
2. **Ramsey solution:** maximize social welfare based on  $N_y$  equations and  $N_y + N_z$  control variables.
3. **Dynamic programming:** use Bellman equation and value function iterations or projections possibly with fancy neural networks.

## Control Problem: model summary (a recap)

- **Exogenous variables** each driven by an exogenous process (growth/decline path).

$$x_t = \{A_t, \sigma_t, \theta_{1,t}, F_t^{EX}, E_t^{\text{land}}, L_t\}$$

- **Endogenous state/flow variables**

$$y_t = \{M_t^{AT}, M_t^{UP}, M_t^{LO}, T_t^{AT}, T_t^{LO}, Y_t, K_t, C_t\}$$

- **Decision variables (controls)**

$$z_t = \{\mu_t, s_t\}$$

- **Horizon and time step:**  $t \in \{2015, 2020, 2025, \dots, T\}$ ,  $\Delta = 5$  years (quinquennial).

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## Control Problem: a sketch of the numerical solution

- Because the social welfare is forward looking, the system is both forward and backward looking as follows:

$$z_t = \arg \min_{\{s_t, \mu_t\}_{i \geq 0}} - \sum_{i=0}^{\infty} (1 + \rho)^{-\Delta i} L_{t+i} u(c_{t+i})$$

- When a sequence  $\{z_{t+i}\}_{i=0}^{\infty}$  is found, one can compute:

$$\begin{aligned} x_t &= g(x_{t-1}), & g : \mathbb{R}^{N_x} &\rightarrow \mathbb{R}^{N_x}, \\ y_t &= f(y_{t-1}, x_t, z_t), & f : \mathbb{R}^{N_y} \times \mathbb{R}^{N_x} \times \mathbb{R}^2 &\rightarrow \mathbb{R}^{N_y}. \end{aligned}$$

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## Control Problem: a sketch of the numerical solution

- **Idea.** Approximate the **infinite-horizon** objective with a **finite** one. Let  $\beta_{\Delta} \equiv (1 + \rho)^{-\Delta}$ . Because  $\lim_{i \rightarrow \infty} \beta_{\Delta}^i = 0$ , the tail contributes little.
- **Pick the horizon  $I^*$**  so that the tail weight is below a tolerance  $\varepsilon$ :

$$\beta_{\Delta}^{I^*} \leq \varepsilon$$

*Trade-off:* smaller  $I^* \Rightarrow$  faster solve but larger truncation error.

- **Finite-horizon problem** with  $T = t + I^* \Delta$ , solve

$$\max_{\{s_{t+i}, \mu_{t+i}\}_{i=0}^{I^*}} \sum_{i=0}^{I^*} \beta_{\Delta}^i L_{t+i\Delta} u \left( 1000 \times \frac{C_{t+i\Delta}}{L_{t+i\Delta}} \right)$$

s.t. dynamics (resource, capital, emissions, climate) and bounds.

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## Control Problem: Two Policies

- The **mitigation policy**:

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- The *laissez-faire* / Business As Usual (BAU) policy:

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$$s.t. \mu_t = 0.03$$



## Control Problem: Two Policies

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- The *laissez-faire* / Business As Usual (BAU) policy:

$$\max_{\{s_{t+i}\}_{i=0}^{I^*}} \sum_{i=0}^{I^*} \beta_{\Delta}^i L_{t+i\Delta} u\left(1000 \times \frac{C_{t+i\Delta}}{L_{t+i\Delta}}\right)$$

s.t. dynamics (resource, capital, emissions, climate) and bounds.

$$s.t. \mu_t = 0.03$$

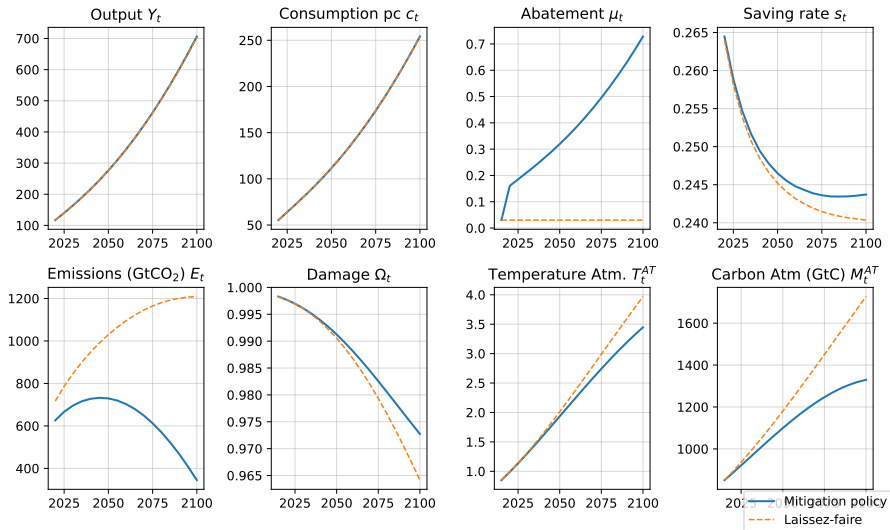


Figure. Simulations under two policy scenarios

# The SSC in the Ramsey Problem

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- Why is it optimal to warm so much the planet according to DICE?
- An optimal carbon tax = how much the society is willing to pay to reduce emissions (\$ per ton of carbon).
- This is usually referred to as the Social Cost of Carbon (SCC) in literature.
- SCC reflects the society's gains and losses from implementing the carbon tax.
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The **Ramsey (1927)** problem:

$$\begin{aligned}
 & \max_{\{c_t, Y_t, K_t, T_t, M_t, \mu_t, s_t\}} \sum_{i=0}^{\infty} \beta_{\Delta}^i L_{t+i} u(c_{t+i}) \\
 & + \beta_{\Delta}^i \lambda_{1,t+i} \left[ (1 - s_{t+i}) Y_{t+i} \left( 1 - \theta_{1,t+i} \mu_{t+i}^{\theta_2} \right) \Omega(T_{t+i-1}) - L_{t+i} c_{t+i} / 1,000 \right] \\
 & + \beta_{\Delta}^i \lambda_{2,t+i} \left[ K_{t+i} - (1 - \delta_K) K_{t+i-1} - s_{t+i} Y_{t+i} \left( 1 - \theta_{1,t+i} \mu_{t+i}^{\theta_2} \right) \Omega(T_{t+i-1}) \right] \\
 & + \beta_{\Delta}^i \lambda_{3,t+i} \left[ Y_{t+i} - A_{t+i} K_{t+i-1}^{\gamma} L_{t+i}^{1-\gamma} \right] \\
 & + \beta_{\Delta}^i \lambda_{4,t+i} \left[ T_{t+i} - \Phi_T T_{t+i-1} - \Xi_T F_{t+i}(M_{t+i}) \right] \\
 & + \beta_{\Delta}^i \lambda_{5,t+i} \left[ M_{t+i} - \Phi_M M_{t+i-1} - \Xi_M \left( \sigma_{t+i} (1 - \mu_{t+i}) Y_{t+i} + E_{t+i}^{\text{land}} \right) \right]
 \end{aligned}$$

**Summary:** optimal control problem with 7 control variables, 5 constraints  $\rightarrow$  12 variables/equations.

(note that climate variables include more equations + Lagrangian multipliers)

- Recall that compact form for climate block with

$$T_t = [T_t^{AT}, T_t^{LO}]^\top \text{ and } M_t = [M_t^{AT}, M_t^{UP}, M_t^{LO}]^\top$$

- The corresponding multipliers:

$$\lambda_{4,t} = [\lambda_{5,t}^{AT}, \lambda_{5,t}^{LO}]^\top \text{ and } \lambda_{5,t} = [\lambda_{5,t}^{AT}, \lambda_{5,t}^{UP}, \lambda_{5,t}^{LO}]^\top$$

- Damages and forcing functions as function of vectors  $T_t$  and  $M_t$  :

$$\Omega(T_t) = 1 / \left( 1 + a_2 \left( T_t^{AT} \right)^2 \right)$$

$$F_t(M_t) = \eta \log \left( M_t^{AT} / M_{1750} \right) / \log(2) + F_t^{EX}$$

- Corresponding gradients:

$$\nabla_T \Omega(T_t) = \left[ -\Omega(T_t)^2 2a_2 \left( T_t^{AT} \right) \quad 0 \right]^\top$$

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- Get back to FOCs of the Ramsey-planner:

$$T_t : \lambda_{4,t} = \beta_{\Delta} \left[ \Phi_T^{\top} \lambda_{4,t+1} + \lambda_{2,t+1} s_{t+1} Y_{t+1} \left( 1 - \theta_{1,t+1} \mu_{t+1}^{\theta_2} \right) \nabla_T \Omega(T_t) \right]$$

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$\lambda_{5,t}$  ( $\lambda_{4,t}$ ) is the marginal loss from carbon (temperature) increase in each climate box.

- SCC expresses the social loss into numeraire equivalents (here consumption):

$$SSC_t = -1000 \times \lambda_{5,t}^{AT} / \lambda_{1,t} \simeq -1000 \times \left( \partial W_t / \partial M_t^{AT} \right) / \left( \partial W_t / \partial C_t \right)$$

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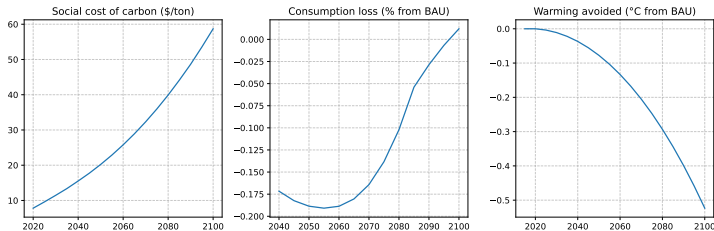
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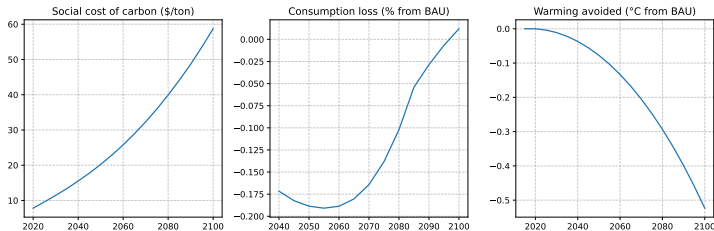
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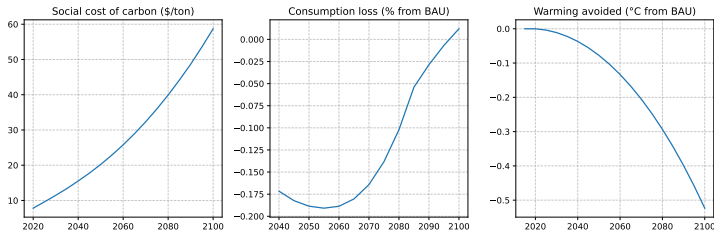


- SCC reflect the planner's relative gains in terms of welfare from controlling  $M_t$  against its consumption losses  $C_t$  from such action;
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- How to retrieve the SCC if we don't have any lagrangian multiplier?
- One can use the firm's static problem (given output  $Y_t$ ):

$$\min_{\mu_t \in [0,1]} \underbrace{\theta_{1,t} \mu_t^{\theta_2} Y_t}_{\text{abatement cost}} + \tau_t \underbrace{\sigma_t (1 - \mu_t) Y_t}_{\text{taxed residual emissions}}.$$

- First-order condition in  $\mu_t$ :

$$\theta_2 \theta_{1,t} \mu_t^{\theta_2-1} Y_t = \tau_t \sigma_t Y_t \quad \Rightarrow \quad \boxed{\tau_t = \frac{\theta_2 \theta_{1,t}}{\sigma_t} \mu_t^{\theta_2-1}}$$

- This tax is in trillion USD per gigatonne carbon. The carbon tax  $\tau_t^\$$  in  $\text{US\$}/\text{tCO}_2$ :

$$\tau_t^\$ = \frac{10^{12}}{10^9} \times \tau_t = 1,000 \times \tau_t$$

as  $1\text{GtC} = 10^9$  and 1 *trillion* \$ =  $10^{12}$ .

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## Recap

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- **Equations:** You don't need to memorize every equation by heart. But you should be able to recognize them if you see them, and explain their meaning.
- **Simulations:** You don't need to know the code. What matters is that you understand the outputs (graphs, tables, numbers) and can interpret them correctly.
- **MCQs:** Expect multiple-choice questions with text, equations, and figures. The goal is to test whether you can read, interpret, and choose the correct answer.

In short: focus on understanding, not rote learning.



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Thank you!

`gauthier@vermandel.fr`

## References

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